

Khetigadi: System Architecture Document

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May 22, 2026

Abstract

This document presents the comprehensive system architecture for Khetigadi, India's asset-light farm mechanization platform. The architecture is designed to support 8.5 million active farmers, 25,000 Mitra agents, 50,000+ telematics-enabled equipment units, and Rs.2,500+ crore in annual revenue by FY30F. The architecture follows a microservices-based, cloud-native design with offline-first capabilities for rural connectivity. Key components include: (1) multi-channel farmer booking system, (2) telematics and IoT device management, (3) demand-supply matching engine, (4) Mitra management and franchise system, (5) spare parts e-commerce engine, (6) NBFC lending and credit underwriting module, (7) data monetization platform, and (8) analytics and reporting layer. The architecture emphasizes scalability, fault tolerance, security, and compliance with Indian data protection regulations.

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1 Architecture Overview

Khetigadi's system architecture is built on a microservices-based, cloud-native design deployed on AWS (Mumbai region) with edge nodes for offline operation in low-connectivity rural areas. The architecture supports the following key user personas:

- **Farmers:** Book equipment, purchase spare parts, pay for rentals, file insurance claims
- **Mitras (franchise owners):** Onboard farmers, manage local equipment, coordinate maintenance, earn commissions
- **Asset owners:** List equipment, track utilization, receive payments, schedule maintenance
- **Operators:** Receive assignments, navigate to fields, log issues, get rated by farmers
- **Corporate users (OEMs, insurers, FPOs):** Access data APIs, purchase insights, manage fleets
- **Khetigadi admins:** Manage platform, approve Mitras, monitor KPIs, handle disputes

1.1 High-Level Architecture Diagram

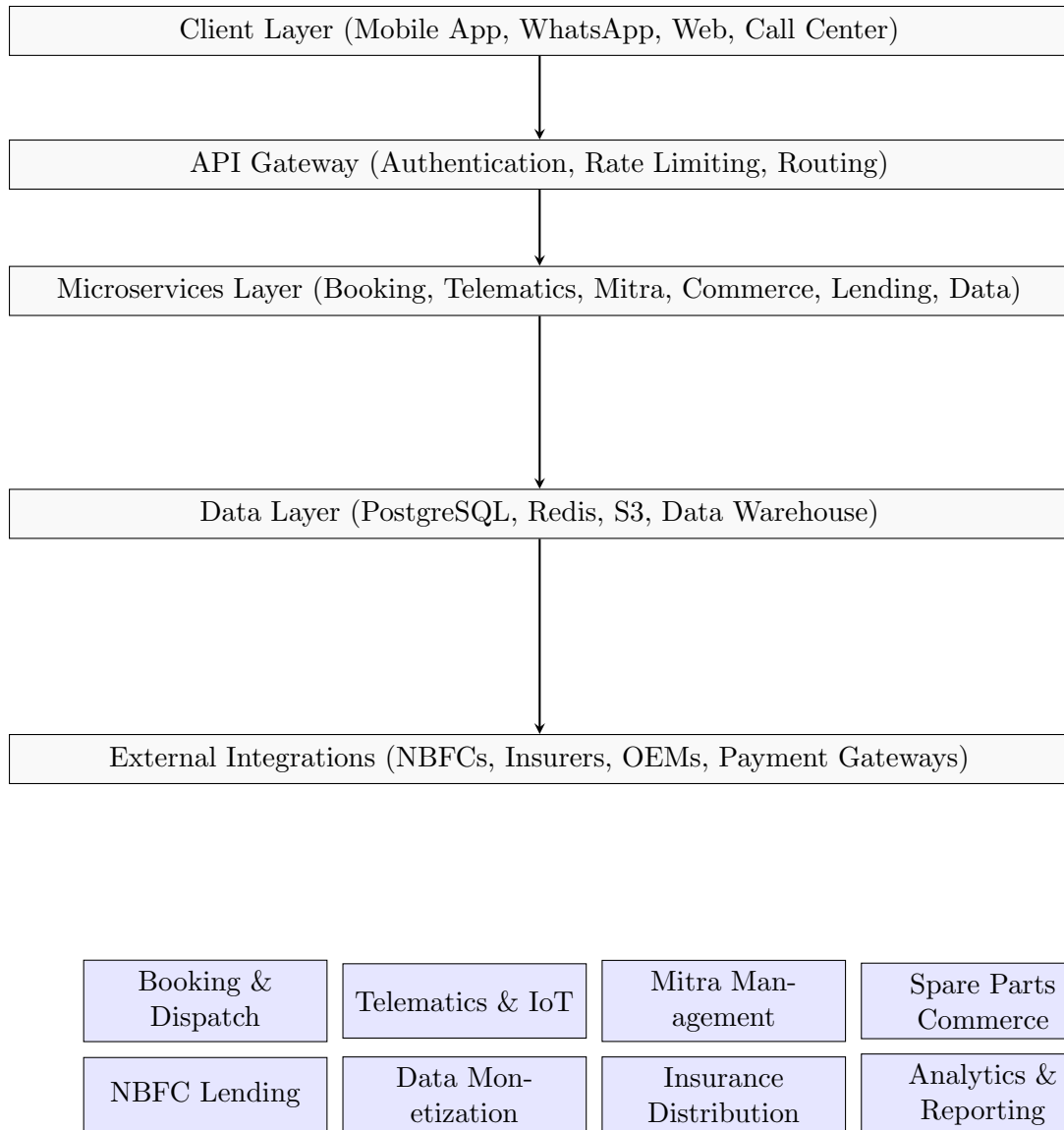


Figure 1: High-Level System Architecture

2 Client Layer

The client layer provides multi-channel access to Khetigadi's services, catering to farmers with varying levels of digital literacy and connectivity.

2.1 Mobile Application (Android & iOS)

- **Technology:** React Native (cross-platform), offline-first with SQLite local storage
- **Key features:**

- Equipment booking (tractor, harvester, implements) with date, duration, and operation type
- Real-time tracking of assigned equipment (GPS location, ETA)
- Spare parts catalog with authentication verification (QR code scan)
- Payment gateway integration (UPI, net banking, cash on delivery)
- Rating and feedback for operators and equipment
- Multilingual support (Hindi, Punjabi, Marathi, Gujarati, Telugu, Kannada, Tamil, English)
- Offline mode: bookings stored locally and synced when connectivity returns
- **Offline capability:** Full booking flow works without internet; data sync via Work-Manager (Android) and BackgroundTasks (iOS)

2.2 WhatsApp Integration

- **Technology:** WhatsApp Business API (Meta)
- **Use cases:**
 - Booking: Farmer sends "BOOK [tractor] [date] [location]" → automated response with availability
 - Order status: Receive notifications for booking confirmation, equipment arrival, payment receipt
 - Support: Raise tickets via WhatsApp; agronomist/mechanic responds within 4 hours
 - Marketing: Broadcast crop-specific offers and maintenance tips
- **Language support:** Natural language processing (NLP) for Hinglish and regional languages

2.3 Web Portal (for Mitras & Corporate Users)

- **Technology:** React.js with TypeScript
- **Mitra portal features:**
 - Farmer onboarding (Aadhaar eKYC, land record upload)
 - Equipment management (add/remove asset owners, schedule maintenance)
 - Commission dashboard (real-time earnings, payout requests)
 - Inventory management (spare parts stock at local hub)
- **Corporate portal features:**
 - API key management
 - Data subscription management
 - Fleet analytics dashboard (for OEMs and FPOs)

2.4 Call Center Integration

- **Technology:** Ozonetel or Exotel (cloud telephony)
- **Features:**
 - IVR (Interactive Voice Response) in 8 regional languages
 - Automated call-back for missed calls (farmer gives missed call → system calls back)
 - Agent dashboard with customer 360° view (booking history, payment status, equipment rented)
 - Screen pop with AI-suggested responses based on farmer’s past interactions

3 API Gateway Layer

The API Gateway serves as the single entry point for all client requests, providing security, routing, and rate limiting.

Table 1: API Gateway Configuration

Component	Specification
Technology	Kong API Gateway / AWS API Gateway
Authentication	JWT (JSON Web Tokens) + OTP for farmers
Rate Limiting	100 requests/minute per farmer; 1,000 req/min per Mitra
Request Routing	Based on endpoint prefix (/booking, /telematics, /commerce, /lending)
SSL/TLS	TLS 1.3 with automatic certificate renewal (Let’s Encrypt)
DDoS Protection	AWS Shield + WAF (Web Application Firewall)
Logging	Centralized logging to ELK stack (Elasticsearch, Logstash, Kibana)

4 Microservices Layer

The microservices layer is the core of Khetigadi’s architecture, comprising loosely coupled services that can be scaled independently.

4.1 Booking & Dispatch Service

This service handles the core equipment rental workflow from search to confirmation.

- **Technology:** Java Spring Boot / Node.js
- **Database:** PostgreSQL (read-write) + Redis (caching)

- **Key functions:**

- **Search:** Find available equipment within 30 km radius, filtered by type, price, operator rating
- **Pricing engine:** Dynamic pricing based on demand (peak season markup 20-30%, off-season discount 10-15%)
- **Dispatch:** Assign nearest available equipment using geospatial query (Post-GIS)
- **Scheduling:** Prevent double-booking; maintain operator availability calendar
- **Payment:** Integrate with Razorpay/PayU; hold payment in escrow until completion
- **Post-rental:** Trigger feedback request, release escrow payment, update operator rating

4.1.1 Demand-Supply Matching Algorithm

The matching algorithm uses a multi-objective optimization approach:

$$\text{Score} = w_1 \cdot \frac{1}{\text{Distance}} + w_2 \cdot \text{OperatorRating} + w_3 \cdot \text{EquipmentReliability} - w_4 \cdot \text{Price}$$

Where: - $w_1 = 0.4$ (distance weight) - $w_2 = 0.3$ (operator rating weight) - $w_3 = 0.2$ (equipment reliability weight from telematics) - $w_4 = 0.1$ (price sensitivity weight)

4.2 Telematics & IoT Service

This service ingests, processes, and analyzes data from GPS devices installed on equipment.

- **Technology:** Golang (high-throughput ingestion) + Apache Kafka (stream processing)
- **Data ingestion:** 500 million data points/day (location, engine hours, fuel level, error codes)
- **Key functions:**
 - **Real-time tracking:** Update equipment location every 30 seconds; expose via WebSocket to farmer app
 - **Predictive maintenance:** Analyze engine hour trends, error code frequency, vibration patterns to predict failures
 - **Fuel monitoring:** Detect unusual fuel consumption (potential theft) using ML anomaly detection

- **Operator behavior scoring:** Monitor sudden braking, overspeeding, idling time; generate safety score (1-100)
- **Geofencing:** Alert if equipment moves outside designated operational area
- **Device management:** OTA (over-the-air) firmware updates; battery monitoring; device health dashboard

4.2.1 Telematics Data Schema

Table 2: Telematics Data Points Collected

Data Field	Frequency	Use Case
GPS location	30 sec	Tracking, geofencing
Engine hours	1 min	Utilization billing, maintenance
Fuel level	5 min	Fuel efficiency, theft detection
Engine RPM	1 min	Operator behavior
Error codes (OBD)	Real-time	Predictive maintenance
Vibration (accelerometer)	10 sec	Equipment health
Battery voltage	5 min	Device health

4.3 Mitra Management Service

This service manages the network of 25,000+ Mitra franchise agents.

- **Technology:** Python Django + Celery (async tasks)
- **Key functions:**
 - **Onboarding:** Aadhaar eKYC, background verification, training certification, agreement signing
 - **Commission calculation:** Real-time accrual of 2-3% on transaction value; monthly payout via UPI
 - **Franchise fee collection:** Integrate with payment gateway; track fee payment status
 - **Performance dashboard:** Leaderboard of top Mitras by GMV, farmer addition, farmer satisfaction
 - **Inventory management:** Track spare parts inventory at each Mitra’s local hub; auto-reorder at threshold
 - **Geo-attendance:** Ensure Mitra is present in assigned territory during working hours

4.4 Spare Parts Commerce Service

This service powers the D2C spare parts brand "Khetigadi Genuine Parts."

- **Technology:** Node.js + Elasticsearch (product search)
- **Key functions:**
 - **Catalog management:** 5,000+ SKUs (filters, belts, oils, bearings, batteries)
 - **Authentication:** QR code on each product; farmer scans to verify authenticity on blockchain
 - **Price engine:** Dynamic pricing based on Mitra zone, volume discounts for FPOs
 - **Last-mile delivery:** Route optimization for Mitra delivery fleet using Google Maps API
 - **Returns management:** Warranty claims (12 months on select parts); return shipping label generation

4.5 NBFC Lending Service

This service provides equipment financing to farmers, underwritten using telematics data.

- **Technology:** Java Spring Boot + Drools (rules engine)
- **Key functions:**
 - **Credit underwriting:** ML model using 50+ features (telematics usage pattern, repayment history, crop type, land holding, seasonal income)
 - **Loan origination:** Digital application with e-sign; Aadhaar-based KYC; disbursement to farmer's bank account
 - **Collections:** Automated post-harvest deduction from output sales (integration with output marketplace)
 - **Delinquency management:** Early warning system using payment behavior and crop health telematics
 - **Regulatory reporting:** Automated RBI (Reserve Bank of India) reporting for NBFC compliance

4.5.1 Credit Scoring Model Features

4.6 Data Monetization Service

This service sells anonymized, aggregated data to enterprise customers via API.

Table 3: Credit Scoring Model Input Features

Category	Features
Telematics	Equipment usage frequency, maintenance compliance, operator rating
Transaction history	Rental frequency, payment timeliness, average order value
Crop data	Crop type, acreage, historical yield (from IoT sensors)
Demographic	Land holding (acres), village, FPO membership
External data	CIBIL score (if available), monsoon forecast for region

- **Technology:** Python FastAPI + Apache Spark (data aggregation)
- **Key functions:**
 - **Data anonymization:** Remove PII (personally identifiable information); aggregate at district/block level
 - **API access:** RESTful APIs with API keys; rate limiting by subscription tier
 - **Report generation:** Scheduled reports (daily, weekly, monthly) for OEMs and government
 - **Data quality monitoring:** Ensure 99.9% data accuracy; alert on anomalies
- **Datasets available for purchase:**
 - Equipment utilization by district, crop, season
 - Maintenance failure patterns by equipment make and model
 - Operator performance benchmarks
 - Spare parts demand forecasting
 - Rental price indices by region

4.7 Insurance Distribution Service

This service enables farmers to purchase equipment insurance through the platform.

- **Technology:** Node.js + integration APIs (AIC, HDFC Ergo, Pramerica)
- **Key functions:**
 - **Quote generation:** Real-time premium calculation based on equipment value, usage hours, operator rating
 - **Policy issuance:** Digital policy document with QR code for verification
 - **Claims processing:** Farmer uploads damage photos; AI assesses damage severity; adjuster assigned for high-value claims
 - **Commission tracking:** Auto-accrual of 8-10% commission; monthly payout to Mitra

4.8 Analytics & Reporting Service

This service provides business intelligence and operational dashboards.

- **Technology:** Apache Superset / Tableau (visualization) + dbt (data transformation)
- **Key dashboards:**
 - **Executive dashboard:** Revenue, GMV, active farmers, LTV:CAC, EBITDA margin by region
 - **Operations dashboard:** Booking fulfillment rate, response time, equipment utilization by hub
 - **Mitra dashboard:** Commission earnings, farmer addition rate, satisfaction score by agent
 - **Telematics dashboard:** Real-time equipment location, maintenance alerts, operator safety scores
 - **Finance dashboard:** Cash flow, working capital, collection efficiency, loan book quality

5 Data Layer

The data layer consists of multiple storage systems optimized for different use cases.

5.1 Operational Databases

Table 4: Operational Databases

Database	Technology	Purpose
Primary OLTP	PostgreSQL 15	Farmers, bookings, payments
Time-series	TimescaleDB	GPS, engine hours, sensors
Geospatial	PostGIS	Distance-based search
Caching	Redis (cluster)	Sessions, rate limiting, availability
Search	Elasticsearch	Parts catalog, equipment search

5.2 Data Warehouse

- **Technology:** Snowflake / Amazon Redshift
- **Data sources:** All microservices stream data to Kafka → transformed via dbt → loaded to warehouse

- **Data volume:** 10 TB raw data per month (projected by FY30F)
- **Refresh frequency:** Real-time (streaming) for operational dashboards; daily for analytical reports

5.3 Data Lake

- **Technology:** Amazon S3 (with Glacier for archival)
- **Content:**
 - Raw telemetry data (retained 90 days in hot storage, 7 years in cold storage)
 - Farmer KYC documents (Aadhaar, land records) - encrypted at rest
 - Equipment inspection photos and videos
 - System logs (retained 30 days)

5.4 Data Governance & Security

- **Encryption:** AES-256 at rest (S3, RDS); TLS 1.3 in transit
- **Access control:** Row-level security in PostgreSQL; IAM roles for AWS services
- **Compliance:** DPDP Act (India), ISO 27001 (planned FY27), SOC 2 Type II (planned FY28)
- **Backup:** Automated daily snapshots; cross-region replication to ap-south-1 (Mumbai) and ap-southeast-1 (Singapore)

6 External Integrations

Khetigadi integrates with several external systems to enable seamless operations.

6.1 Payment Gateways

Table 5: Payment Gateway Integrations

Gateway	Use Case
Razorpay	UPI, credit/debit cards, net banking
PayU	Cash on delivery reconciliation
Cashfree	Payouts to Mitras and asset owners
Bank API (ICICI/HDFC)	NBFC loan disbursement and collection

6.2 Government Systems

- **Aadhaar eKYC:** UIDAI API for farmer identity verification
- **DigiLocker:** Land record verification
- **GST portal:** Invoice validation and e-way bill generation for spare parts
- **PMFBY (crop insurance):** API integration for policy lookup and claim status

6.3 Third-Party Services

- **Google Maps API:** Distance calculation, route optimization, address autocomplete
- **Twilio:** SMS notifications (fallback when WhatsApp unavailable)
- **Sentry:** Error tracking and performance monitoring
- **Datadog:** Infrastructure monitoring and alerting

7 Scalability & Performance

7.1 Scaling Strategy

Table 6: Horizontal Scaling Targets by FY30F

Component	FY24A	FY30F
Active farmers	1.2 M	8.5 M
Daily bookings	15,000	120,000
Telematics data points/day	50 M	500 M
API requests/second (peak)	500	5,000
Mitra agents	8,000	25,000
Equipment units tracked	15,000	50,000+

7.2 Auto-scaling Configuration

- **Kubernetes:** EKS (Amazon Elastic Kubernetes Service) with Horizontal Pod Autoscaler (HPA)
- **Trigger conditions:** CPU $\geq$ 70%, memory $\geq$ 80%, custom metric (requests per second $\geq$ 1,000)
- **Node groups:**
 - Compute-optimized (c5.large): Booking, telematics ingestion

- Memory-optimized (r5.large): Redis caching, PostgreSQL
- GPU-optimized (g4dn.xlarge): ML model inference (credit scoring, predictive maintenance)

7.3 Performance SLAs

Table 7: Performance Service Level Agreements (SLAs)

Service	Target Response Time
Search equipment (p95)	<500 ms
Booking confirmation (p95)	<1 sec
Telematics location update (p95)	<100 ms
Spare parts checkout (p95)	<2 sec
Loan application decision (p95)	<10 sec (automated) / <24 hr (manual review)
API data access (p95)	<1 sec for aggregated data; <5 sec for raw export

8 Deployment Architecture

8.1 AWS Infrastructure

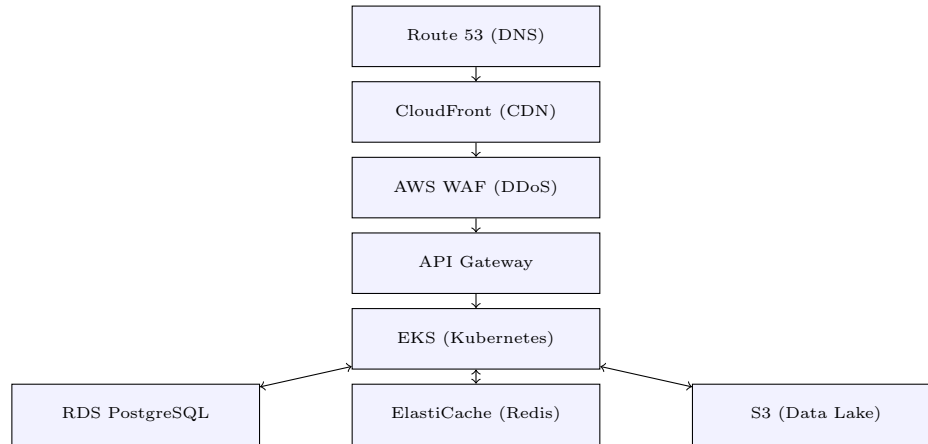


Figure 2: AWS Deployment Architecture

8.2 Multi-Region Strategy

- **Primary region:** ap-south-1 (Mumbai) - all production workloads
- **DR region:** ap-southeast-1 (Singapore) - cross-region replication for critical data (farmer profiles, bookings, payments)
- **Edge locations:** CloudFront CDN with 10+ edge locations in India for static content (app downloads, product images)

8.3 Disaster Recovery

Table 8: Disaster Recovery Objectives

Metric	Target
Recovery Time Objective (RTO)	4 hours
Recovery Point Objective (RPO)	15 minutes (critical data: 5 minutes)
Backup frequency	Daily automated snapshots
DR testing frequency	Quarterly (simulated region failover)

9 Offline-First Design

Given that 30-40% of farmers operate in areas with intermittent connectivity, Khetigadi implements offline-first design principles.

9.1 Offline Capabilities by Component

Table 9: Offline Functionality

Component	Offline	Sync Strategy
Equipment search	Yes	Daily sync
Create booking	Yes	Local SQLite → sync online
Booking history	Yes	Cache after first load
Payment	No	N/A (UPI required)
Scan spare parts QR	Yes	Batch sync
Insurance claim (photos)	Yes	Upload on connectivity
Mitra earnings dashboard	Yes (cached 24h)	Sync on app open

9.2 Sync Conflict Resolution

- **Last-write-wins:** For non-critical fields (e.g., profile preferences)
- **Server wins:** For financial transactions (booking slots, payments)
- **Manual resolution:** For disputed bookings (operator no-show, equipment damage)
- **Idempotency keys:** Prevent duplicate bookings when retrying failed API calls

10 Security Architecture

10.1 Authentication & Authorization

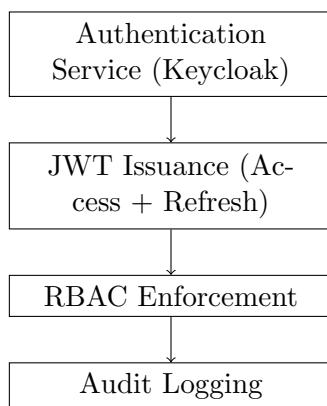


Figure 3: Authentication Flow

10.2 Security Controls

Table 10: Security Controls Summary

Control	Implementation
Authentication	OTP for farmers; JWT for Mitras and corporate users
Authorization	RBAC (Role-Based Access Control) with 8 roles
API Security	JWT validation at API Gateway; rate limiting by role
Data Encryption	AES-256 at rest; TLS 1.3 in transit
Secrets Management	AWS Secrets Manager (API keys, database credentials)
Vulnerability Scanning	Weekly automated scans (AWS Inspector, Snyk)
Penetration Testing	Annual third-party penetration test (CERT-In empaneled)
Audit Logging	All API calls logged with user ID, timestamp, IP address

10.3 GDPR/DPDP Act Compliance

- **Data localization:** All farmer data stored only in India (ap-south-1 region)
- **Consent management:** Explicit consent for data collection; granular opt-out for marketing
- **Data retention:** PII deleted 7 years after last transaction (as per RBI guidelines)
- **Right to be forgotten:** API endpoint for account deletion (90-day processing time)
- **Data breach notification:** 72-hour notification to affected users and CERT-In

11 Monitoring & Observability

11.1 Monitoring Stack

Table 11: Monitoring Tools

Tool	Purpose
Datadog	Infrastructure metrics (CPU, memory, network, disk)
Prometheus + Grafana	App metrics (request rate, error rate, latency)
ELK Stack	Centralized logging
AWS X-Ray	Distributed tracing
Sentry	Error tracking & crash reporting
PagerDuty	On-call alerting

11.2 Key Metrics to Monitor

- **Business KPIs:**
 - GMV per day, week, month (by region, by equipment type)
 - Booking fulfillment rate (booked vs. completed)
 - Average response time (booking to equipment arrival)
 - Mitra commission accrual and payout status
- **Technical KPIs:**
 - API p95 latency by endpoint
 - Error rate (HTTP 5xx) - target $\leq 0.1\%$
 - Telematics data ingestion lag - target ≤ 5 seconds
 - Database connection pool utilization

12 Cost Optimization

12.1 Current & Projected Cloud Spend

12.2 Optimization Strategies

1. **Reserved Instances:** Purchase 3-year reserved instances for stable workloads (30-40% savings)
2. **Spot instances:** Use spot instances for non-production and batch processing (60-70% savings)

Table 12: AWS Monthly Cost Projection (Rs. Lakhs)

Service	FY25F	FY27F	FY30F
Compute (EKS, EC2)	8	15	35
Database (RDS, ElastiCache)	4	8	18
Storage (S3, EBS)	2	5	15
Data transfer (CloudFront, NAT)	3	6	12
Monitoring & security	1	2	5
Total Monthly	18	36	85
Total Annual	216	432	1,020

3. **S3 lifecycle policies:** Transition older telematics data to Glacier after 90 days (80% storage cost reduction)
4. **CloudFront caching:** Cache product images and static assets at edge (70% reduction in origin fetches)
5. **Data compression:** Compress telematics data before transmission (50% reduction in data transfer costs)

13 Conclusion

The Khetigadi system architecture is designed to support the company’s ambitious growth targets from Rs.45 crore (FY24A) to Rs.2,500+ crore (FY30F) while maintaining high availability (99.9% uptime target), sub-second response times, and full compliance with Indian data protection regulations. The microservices-based, cloud-native design enables independent scaling of high-traffic components (booking, telematics), while offline-first capabilities ensure usability in low-connectivity rural areas. The architecture’s modularity also allows for rapid integration of new revenue streams (NBFC lending, data monetization, insurance distribution) as outlined in the Revenue Expansion Playbook.

Key architectural decisions; including the choice of PostgreSQL with TimescaleDB for telematics, Kubernetes for orchestration, and offline-first mobile apps; reflect the unique constraints of serving semi-literate, low-connectivity rural farmers while managing high-velocity telematics data at scale. The architecture is projected to handle 500 million telematics data points per day, 120,000 daily bookings, and 8.5 million active farmers by FY30F.

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